

Black-Scholes Stock Options Pricing Model

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1 Introduction

In this paper I will be going through the process I went through to build my own Black-Scholes stock options pricing model. I begin with building a fundamental understanding of what stock options are before moving onto the quantitative side of option pricing models and then running the model with real data to assess the accuracy.

2 What are Options?

Options are a contract that gives a buyer the right, though not the obligation, to buy or sell an underlying asset at a set price within a specific time-frame. Contracts where the underlying asset is being bought are called call options while those being sold are put options. To help understand this, put yourself in the position of an investor, approaching the purchase of a stock. Suppose the stock-price of *Younus Technologies* lies at \$40 and you project its value to increase after a portion of time, you may buy a call option, paying a premium of \$5 to enter the contract. Let us suppose that the strike price is \$50 and that you choose to exercise your contract when the market value of the stock rises to \$60. Congratulations you just made \$5 in profit. That \$50 strike price would remain constant regardless of whether the stock price rose to \$80 or \$100 etc. meaning that there is a no limit to potential profit but a cap at loss of just the premium price - should you refuse to exercise the contract. This is an example of a long call option.

2.1 Long Calls

In the overview we looked at a scenario where an investor bought and exercised a long call option. From Figure 1 we can make the observation that risk is maximised at the premium but to at least break even, the stock must rise enough to cancel the strike price and the premium.

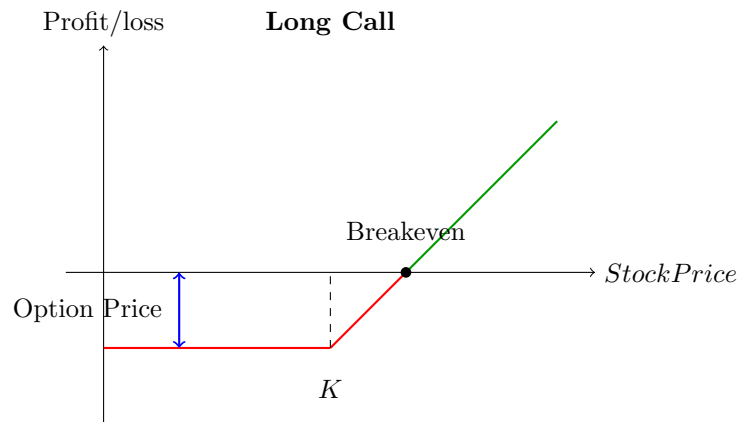


Figure 1: Long Call payoff diagram

2.2 Long Puts

With Long Puts, you are buying the contract expecting the stock price to reduce. Figure 2 illustrates yet again that your maximum loss is the option price. As stock price decreases, your profit increases; the inverse of a Long Call.

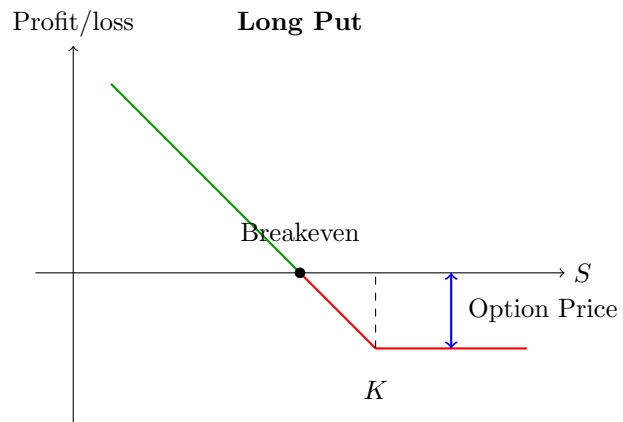


Figure 2: Long Put payoff diagram

2.3 Short Calls

In this scenario you are the one selling the call option contract, collecting the premium and hoping the stock remains level in market price or stays below the predetermined strike price such that the contract expires without being exercised. The loss is unlimited.

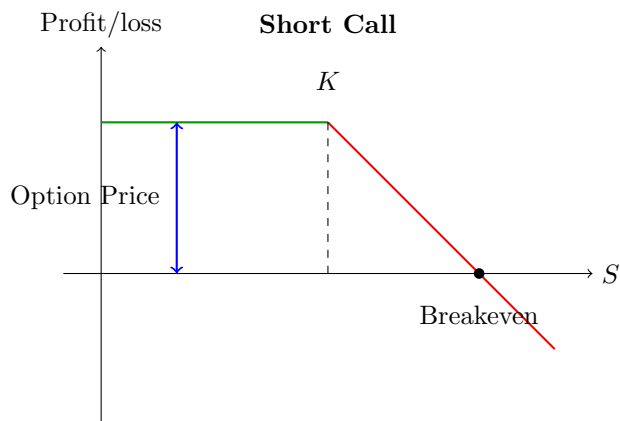


Figure 3: Short Call payoff diagram

2.4 Short Puts

Logically following from the last, here a put option is being sold. The strategy is self-explanatory. The maximum loss is K ; the premium.

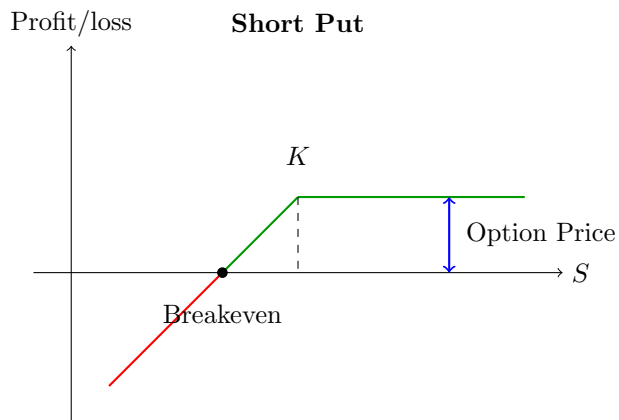


Figure 4: Short Put payoff diagram

3 The Mathematics

3.1 The Equation

$$C = SN(d_1) - Ke^{-rt}N(d_2) \quad (1)$$

- C = Call Option Price
- S = Current Stock Price
- K = Strike Price
- r = Risk-Free Interest Rate
- t = Time to Maturity
- N = Cumulative Distribution Function of the Standard Normal Distribution

The $N(x)$ functions refer to the standardisation normalisation equations that convert inputs into specific points along the Normal Bell Curve. When d_1 and d_2 are substituted into an $N(x)$, they output probabilities. This equation tells the price a stock option should trade for.

3.2 Assumptions

- Volatility is constant and known throughout the contract
- Price follows Brownian Motion
- European Option contract

There are more assumptions made but for the sake of simplicity we shall leave our list here.

3.3 Derivation

In this section we will cover the mathematical derivation of the Black Scholes equation. It is worth going over the mathematics in the reference page to get a good grasp of what is happening.

We begin with one of our assumptions; that the stock price follows geometric Brownian Motion. Brownian Motion is a stochastic process, hence a stochastic differential equation is the base of our derivation.

$$dS = \mu S dt + \sigma S dW \quad (2)$$

The first term is our drift component and the second is our noise component. S is a function telling us the asset price at time t and, since both terms have a factor of S , our output movements are in terms of percentages.

Performing a Taylor expansion on dS gives us dV , the change in the resultant option price after a change in the stock price. The process may feel hard to follow but lets try to approach it using very basic mathematics as a guide.

Suppose first principles of differentiation. We will use this to help us conceptually understand the process.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (3)$$

Firstly, what are we aiming to extract? We want to be able to find what the change in $V(S,t)$ is when S moves a little and when t moves a little. Notation-wise, we will write those as ΔS and Δt . Expressing this as an equation:

$$\Delta V = V(S + \Delta S, t + \Delta t) - V(S, t) \quad (4)$$

Similar to the first principles of differentiation, we are finding an increment change. Dissimilarly, there is no limit to that change as we want our terms to be giving a continuous and accurate correction output.

To progress, we will convert this into a single variable problem, switching back after expansion.

Envision point (S,t) moving towards $(S+\Delta S, t + \Delta t)$. We will use a single variable, h that progresses from start to end; 0 to 1.

$$S(h) = S + h\Delta S \quad (5)$$

$$t(h) = t + h\Delta t \quad (6)$$

Now define a new function of solely h .

$$g(h) = V(S(h), t(h)) \quad (7)$$

- Start: $g(0) = V(S, t)$
- End: $g(1) = V(S + \Delta S, t + \Delta t)$

Therefore,

$$\Delta V = g(1) - g(0) \quad (8)$$

Performing the expansion around h ,

$$\Delta V = \frac{dg}{dh} \Big|_{h=0} + \frac{1}{2} \frac{d^2g}{d^2h} \Big|_{h=0} + \dots \quad (9)$$

Now we find what each derivative with the multivariable chain rule.

First derivative,

$$\frac{dg}{dh} = \frac{\partial V}{\partial S} \cdot \frac{dS(h)}{dh} + \frac{\partial V}{\partial t} \cdot \frac{dt(h)}{dh} \quad (10)$$

$$\frac{dg}{dh} = \frac{\partial V}{\partial S} \cdot \Delta S + \frac{\partial V}{\partial t} \cdot \Delta t \quad (11)$$

Second derivative,

$$\frac{d^2g}{d^2h} = \Delta S \frac{d}{dh} \left(\frac{\partial V}{\partial S} \right) + \Delta t \frac{d}{dh} \left(\frac{\partial V}{\partial t} \right) \quad (12)$$

$$\frac{d^2g}{d^2h} = \Delta S \left(\frac{\partial^2 V}{\partial^2 S} \Delta S + \frac{\partial^2 V}{\partial^2 t \partial S} \Delta t \right) + \Delta t \left(\frac{\partial^2 V}{\partial S \partial t} \Delta S + \frac{\partial^2 V}{\partial^2 t} \Delta t \right) \quad (13)$$

$$\frac{d^2g}{d^2h} = \frac{\partial^2 V}{\partial^2 S} (\Delta S)^2 + \frac{\partial^2 V}{\partial^2 t \partial S} \Delta S \Delta t + \frac{\partial^2 V}{\partial S \partial t} \Delta t \Delta S + \frac{\partial^2 V}{\partial^2 t} (\Delta t)^2 \quad (14)$$

Applying Schwarz's theorem,

$$\frac{d^2g}{d^2h} = \frac{\partial^2 V}{\partial^2 S} (\Delta S)^2 + 2 \frac{\partial^2 V}{\partial^2 t \partial S} \Delta S \Delta t + \frac{\partial^2 V}{\partial^2 t} (\Delta t)^2 \quad (15)$$

Subbing the derivatives into the expansion,

$$\Delta V = \frac{\partial V}{\partial t} \Delta t + \frac{\partial V}{\partial S} \Delta S + \frac{1}{2} \left(\frac{\partial^2 V}{\partial^2 S} (\Delta S)^2 + 2 \frac{\partial^2 V}{\partial^2 t \partial S} \Delta S \Delta t + \frac{\partial^2 V}{\partial^2 t} (\Delta t)^2 \right) + \dots \quad (16)$$

Simplifying and looking at first few terms,

$$\Delta V = \frac{\partial V}{\partial t} \Delta t + \frac{\partial V}{\partial S} \Delta S + \frac{1}{2} \frac{\partial^2 V}{\partial^2 S} (\Delta S)^2 + \frac{\partial^2 V}{\partial^2 t \partial S} \Delta S \Delta t + \frac{1}{2} \frac{\partial^2 V}{\partial^2 t} (\Delta t)^2 \quad (17)$$

Adjusting notation,

$$dV \approx \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial^2 S} (dS)^2 + \frac{\partial^2 V}{\partial^2 t \partial S} dS dt + \frac{1}{2} \frac{\partial^2 V}{\partial^2 t} (dt)^2 \quad (18)$$

Going back to the Brownian Motion equation,

$$dS = \mu S dt + \sigma S dW \quad (19)$$

$$(dS)^2 = \mu^2 S^2 (dt)^2 + 2\mu\sigma S^2 dt dW + \sigma^2 S^2 (dW)^2 \quad (20)$$

We drop the $(dt)^2$ and $dt dW$ terms as $dW \sim \sqrt{dt}$ so $dt \cdot dW$ and $(dt)^2$ vanish faster than dt

$$(dS)^2 = \sigma^2 S^2 (dW)^2 \quad (21)$$

We move to substituting our expressions into our Taylor expansion.

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial^2 S} (dS)^2 \quad (22)$$

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} (\mu S dt + \sigma S dW) + \frac{1}{2} \frac{\partial^2 V}{\partial^2 S} (\sigma^2 S^2 (dW)^2) \quad (23)$$

$$(dW)^2 = dt \quad (24)$$

Where does (24) come from? $(dW)^2$ is random but it has a mean dt and a variance $2(dt)^2$. It can be said that its order is dt but its randomness is order $(dt)^2$ which is much smaller. Consider the following expression where n small steps of length $\frac{T}{n}$ are summed over a time interval T .

$$\sum_{i=1}^n (\Delta W_i)^2 \quad (25)$$

(25) has a mean T and a variance $\frac{2T^2}{n}$. As $n \rightarrow \infty$ the variance tends to 0, leaving the sum to become exactly T . Therefore, we can take (24) because we are summing over time; essentially what an equation in dt means.

Let's get back to our equation and tidy up.

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S} \right) dt + \sigma S \frac{\partial V}{\partial S} dW \quad (26)$$

Let's build a portfolio from one option and a short position of Δ shares. This is called delta hedging; making the portfolio riskless; as per one of the assumptions.

$$\Pi = V - \Delta S \quad (27)$$

$$d\Pi = dV - \Delta dS \quad (28)$$

Substituting,

$$d\Pi = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S} \right) dt + \sigma S \frac{\partial V}{\partial S} dW - \Delta (\mu S dt + \sigma S dW) \quad (29)$$

$$d\Pi = \left(\frac{\partial V}{\partial t} + \mu S \left(\frac{\partial V}{\partial S} - \Delta \right) + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S} \right) dt + \sigma S \left(\frac{\partial V}{\partial S} - \Delta \right) dW \quad (30)$$

Let,

$$\Delta = \frac{\partial V}{\partial S} \quad (31)$$

As a result, the dW and μ terms vanish,

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt \quad (32)$$

Our portfolio is now riskless. As an example, if the option gains \$0.67 when the stock rises by \$1, you short 0.67 shares so both moves are offset.

If Π earned more than r then you could borrow at r and buy Π for guaranteed profit. If less was earned, the reverse would happen. With no arbitrage,

$$d\Pi = r\Pi dt = r \left(V - S \frac{\partial V}{\partial S} \right) dt \quad (33)$$

Equate (32) and (33), cancelling the dt ,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV - rS \frac{\partial V}{\partial S} \quad (34)$$

Moving things around a little,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (35)$$

We have our time decay balanced against the curvature. In hedged positions, the equation tells us that the loss from time passing, $\frac{\partial V}{\partial t}$, and the gain from price swings, $\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}$ equals the financing cost; traders call this the gamma-theta trade-off.

μ is no longer present and the option can be built from any stock so any view on the stock's growth is as a result seen in its price. You get the same number as in a world where every asset grows at r . This is risk-neutral pricing.

I'll let the reader take the time to complete the last stages of the proof as, at this stage, my insight into the mathematics required is being eclipsed. Refer to the references document for pointers in the right direction. The final equation is the solution of (35) with terminal condition:

$$C(S, T) = \max(S_T - K, 0) \quad (36)$$

Finally,

$$C = SN(d_1) - Ke^{-rt}N(d_2) \quad (37)$$

I will define d_1 and d_2 but without further derivation or definition; consult references if interested in further reading.

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} \quad (38)$$

$$d_2 = d_1 - \sigma\sqrt{t} \quad (39)$$

4 Pricing Options: The Black–Scholes Model

This section explains the model, then shows how it was implemented in Python, how the implementation was verified, and how well it prices real Apple (AAPL) options.

4.1 From payoff to price

The premium has two parts. The intrinsic value is what the option would pay if it were to be exercised now. The time value is everything else; it pays for the chance that the stock moves further in the holder's favour before expiry. For an AAPL call with strike $K = \$340$, with the stock at $S = \$341.07$ and 28 days to expiry, the intrinsic value is \$1.07 and the Black–Scholes price is \$9.02, so the time value is \$7.95.

As expiry approaches the time value decays (Table 1), and the smooth price curve sinks onto the payoff diagram (Figure 5).

The price depends on six inputs, listed in Table 2. Every input except volatility can be looked up. Volatility has to be estimated, which is why it sits at the centre of the results that follow.

4.2 The Greeks

The Greeks are partial derivatives of the option price with respect to its inputs. They measure what a position is exposed to and how to hedge it. Table 3 gives their values for the $K = 340$ options and Figure 6 shows how they vary with the stock price.

Table 1: Time decay of the $K = 340$ AAPL call with the stock held at \$341.07 ($\sigma = 21\%$, $r = 4.22\%$).

Days to expiry	Price	Intrinsic value	Time value
28	\$9.02	\$1.07	\$7.95
14	\$6.43	\$1.07	\$5.36
7	\$4.66	\$1.07	\$3.59
1	\$2.11	\$1.07	\$1.04
0 (expiry)	\$1.07	\$1.07	\$0.00

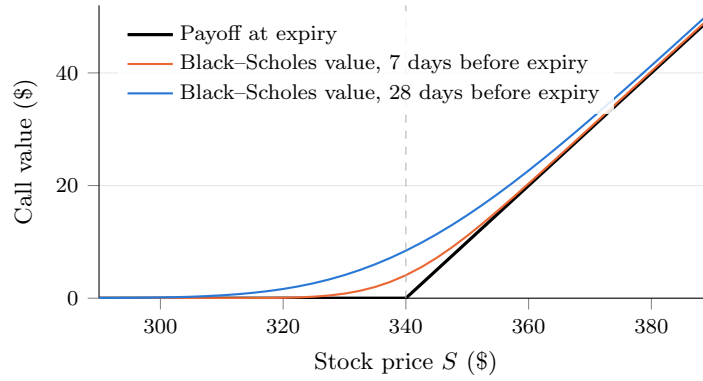


Figure 5: The Black-Scholes value of the $K = 340$ call sits above the payoff diagram and sinks onto it as expiry approaches. The gap between the curve and the payoff is the time value.

Table 2: The inputs to the Black-Scholes model, with the values used for AAPL.

Input	Symbol	AAPL value	Source
Stock price	S	\$341.07	Closing price, 25 Sep 2026
Strike	K	\$240–\$440	The option contract
Time to expiry	T	28 or 84 days	Calendar days $\div$ 365
Risk-free rate	r	4.22%	3-month US Treasury yield
Volatility	σ	20–37%	Estimated
Dividends	q or D	\$0.27 before December	Apple’s quarterly dividend

Table 3: Greeks for the $K = 340$ call and put ($S = 341.07$, 28 days, $\sigma = 21\%$, $r = 4.22\%$).

Greek	Definition	Call	Put	Units
Delta, Δ	$\partial V / \partial S$	0.555	−0.445	per \$1 move in S
Gamma, Γ	$\partial^2 V / \partial S^2$	0.0199	0.0199	per \$1 move in S
Vega, ν	$\partial V / \partial \sigma$	0.373	0.373	\$ per volatility point
Theta, Θ	$\partial V / \partial t$	−0.161	−0.122	\$ per day
Rho, ρ	$\partial V / \partial r$	0.138	−0.122	\$ per 1% rate move

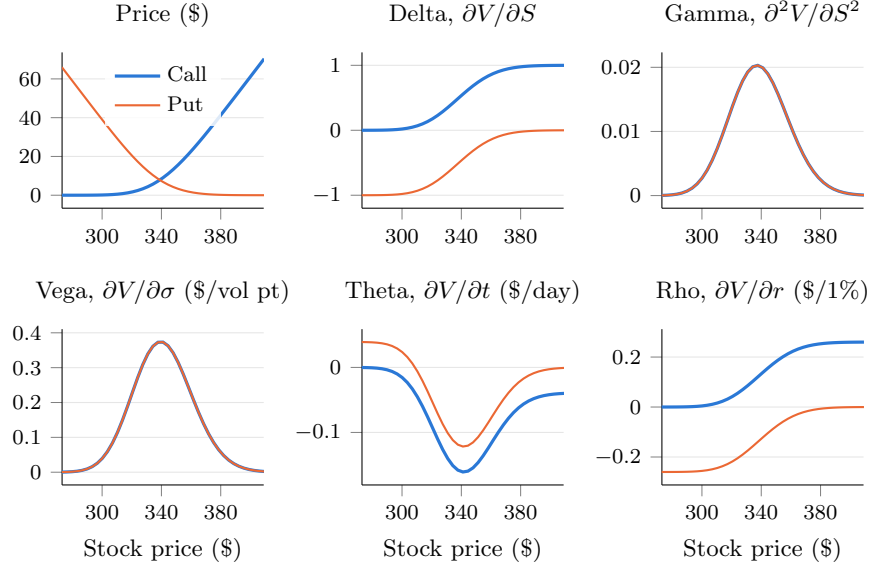


Figure 6: Price and Greeks of the $K = 340$ call (blue) and put (orange), 28 days to expiry, $\sigma = 21\%$, against the stock price. Gamma and vega are identical for calls and puts, so the lines overlap.

The Greeks connect directly to the payoff diagrams:

- Delta is the slope of the price curve
- Gamma is the curvature; concentrated at the strike
- Long and short positions are mirror images. A long option has positive gamma and negative theta; the holder pays time decay for the chance of a large move.

5 Empirical Test on AAPL Options

5.1 Data

The test uses AAPL options at the market close on Friday 25 September 2026 (Table 4). AAPL options are among the most heavily traded single-stock options, so their quotes are reliable. Two expiries were chosen. The first, 23 October 2026, falls before Apple's late-October earnings announcement, which can cause a jump the model does not allow for. The second, 18 December 2026, spans both the earnings announcement and a dividend.

- Option quotes: Cboe delayed quotes, https://cdn.cboe.com/api/global/delayed_quotes/options/AAPL.json

Table 4: Market inputs for the AAPL test, as at the close on 25 Sep 2026.

Input	Value
Spot price S	\$341.07
Risk-free rate r	4.22% continuous (3-month Treasury yield 4.24%)
Historical volatility	24.6% (252 trading days of daily log returns)
Dividends before expiry	\$0.27 ex-dividend 9 Nov 2026 (estimated)
Expiries	23 Oct 2026 (28 days) 18 Dec 2026 (84 days)
Options tested	60 of 131 quotes: out of the money, bid $\geq$ \$0.05, open interest $\geq$ 100

- Daily prices: Cboe historical data, https://cdn.cboe.com/api/global/delayed_quotes/charts/historical/AAPL.json
- Risk-free rate: FRED series DGS3MO, <https://fred.stlouisfed.org/series/DGS3MO>
- Dividends: <https://stockanalysis.com/stocks/aapl/dividend/>

5.2 Method

Because the formula is only as good as the volatility it is given, it was run in two versions:

- **Model A** uses the 1-year historical volatility of 24.6%. This is a genuine forecast, using no information from the options market.
- **Model B** uses the market’s at-the-money implied volatility for each expiry (20.9% for October, 24.3% for December) at every strike. This is Black–Scholes at its best: any error that remains is due to the assumption of constant volatility.

Accuracy is measured by the mean absolute error in dollars, the median absolute percentage error, and the share of options whose model price lies inside the bid–ask spread. With spreads of only \$0.05 to \$0.25, the latter is a stricter test.

5.3 Results

Table 5 summarises the results. Even at its best (Model B), Black–Scholes misses October options by \$0.27 on average and prices only 17% of them inside the bid–ask spread. With historical volatility (Model A) the average error rises to \$0.61.

The errors are not random (Figure 7). Model A overprices the October options, by up to \$1.63 on the $K = 345$ call, because its volatility of 24.6% exceeds the roughly 21% the market expects over the following four weeks. Model B underprices puts below the stock price, by up to \$0.57 in October

Table 5: Black–Scholes prices against market mid-prices for 23 (October) and 37 (December) liquid out-of-the-money AAPL options at the close on 25 Sep 2026. Model A uses 1-year historical volatility; Model B uses each expiry’s at-the-money implied volatility at every strike.

Expiry	Model	σ	Mean abs. error	Median % error	Inside bid–ask
23 Oct (28 days)	A: historical	24.6%	\$0.61	54%	0%
	B: at-the-money	20.9%	\$0.27	20%	17%
18 Dec (84 days)	A: historical	24.6%	\$0.60	26%	5%
	B: at-the-money	24.3%	\$0.59	22%	5%

and \$1.13 in December, and overprices calls above it, by up to \$0.25 and \$0.57 respectively.

Figure 8 shows the pattern. It plots the implied volatility of each option, the σ the market is actually using at each strike. If Black–Scholes were completely accurate, both lines would be flat. Instead, the lower the strike, the higher the volatility the market charges. Model B applies a single flat volatility, so it undercharges at low strikes and overcharges at high ones. The December curve sits higher overall because it covers the earnings announcement.

6 Reflections

Historical volatility is a poor short-term forecast. AAPL moved 24.6% a year over the past year, but its option prices imply only about 21% over the next four weeks. The past year contained several violent days that the market does not expect to repeat soon; the most recent month’s realised volatility, 21.6%, is much closer to the implied figure. This is why practitioners work with implied volatility rather than historical volatility, looking forward rather than back.

No single volatility fits every strike; low-strike puts trade at up to 37% implied volatility, against 20.9–24.3% at the money. Two effects contribute. First, puts act as insurance against a fall in a stock the buyer owns, and demand from investors protecting their portfolios raises their price. Second, real returns have fatter tails than the normal distribution assumes.

6.1 Limitations

- A European formula is applied to American options. The early-exercise effect is measured but not corrected for.
- The test uses a single day of closing quotes. Quotes for thinly traded strikes can be stale.
- Apple’s November dividend had not been announced, so \$0.27 on 9 November 2026 is assumed from its usual pattern.

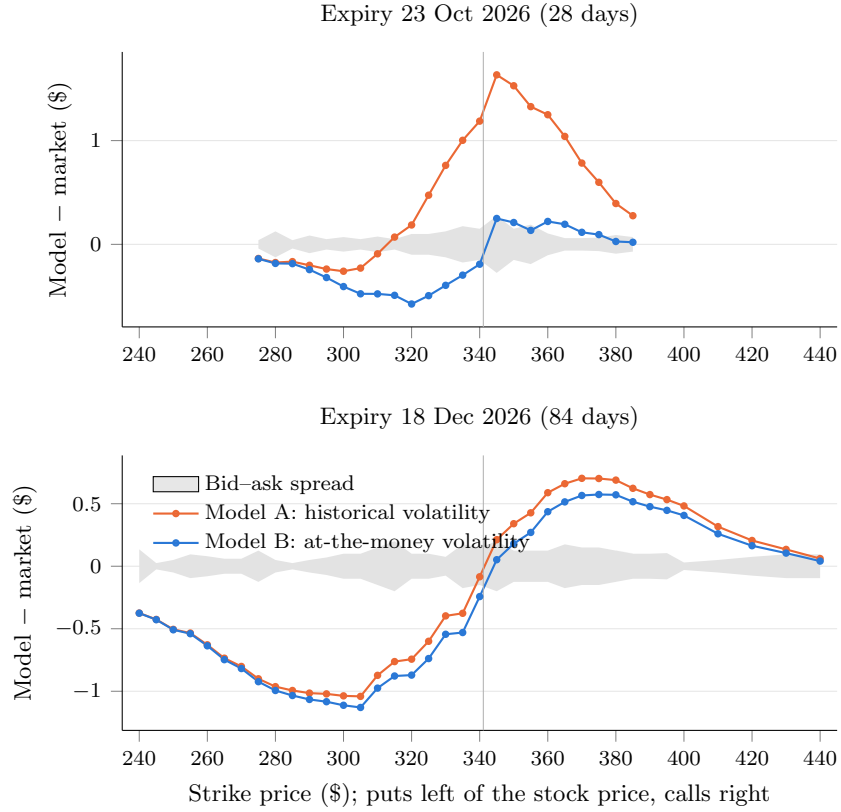


Figure 7: Model price minus market mid-price for every AAPL option tested. Model A uses 24.6% historical volatility; Model B uses the at-the-money implied volatility (20.9% for October, 24.3% for December). Positive values mean the model is too expensive. Inside the grey band the model price lies within the bid-ask spread. The vertical line marks the stock price, \$341.07.

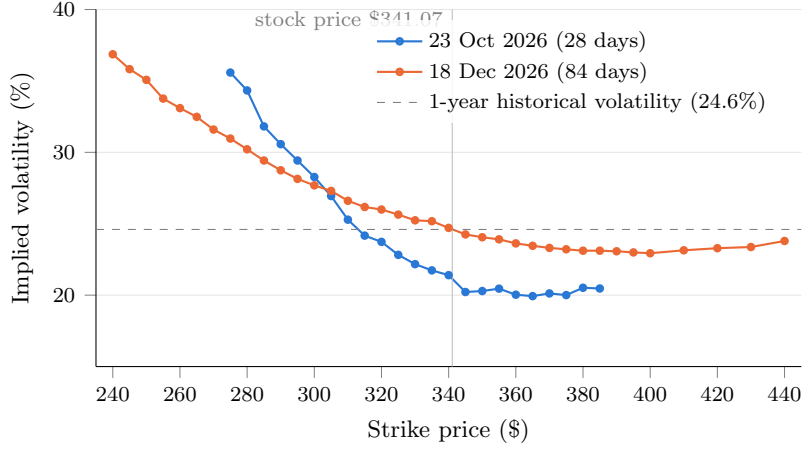


Figure 8: Implied volatility by strike for AAPL options at the close on 25 September 2026. Black–Scholes assumes one volatility for every strike, so under the model each line would be flat.

- Interest rates and volatility are held constant over each option’s life, and the model allows no jumps, such as those caused by earnings announcements.
- Time is measured in calendar days despite markets being closed on the weekends.

7 AAPL Option Data

Table 6: Every AAPL option in the test, closing quotes on 25 Sep 2026. Prices in dollars. IV is this project’s implied volatility from the mid-price; Cboe IV is Cboe’s published figure.

Expiry	Type	Strike	Bid	Ask	Mid	IV (%)	Cboe IV (%)	Model A	Model B
23 Oct	Put	275	0.10	0.18	0.140	35.6	35.6	0.004	0.000
23 Oct	Put	280	0.06	0.31	0.185	34.3	34.4	0.010	0.001
23 Oct	Put	285	0.15	0.23	0.190	31.8	31.8	0.024	0.004
23 Oct	Put	290	0.17	0.34	0.255	30.6	30.6	0.053	0.011
23 Oct	Put	295	0.30	0.40	0.350	29.4	29.4	0.112	0.030
23 Oct	Put	300	0.41	0.55	0.480	28.3	28.2	0.221	0.073
23 Oct	Put	305	0.59	0.69	0.640	26.9	26.9	0.411	0.163
23 Oct	Put	310	0.74	0.89	0.815	25.3	25.2	0.724	0.337
23 Oct	Put	315	1.09	1.19	1.140	24.2	24.1	1.210	0.649
23 Oct	Put	320	1.64	1.84	1.740	23.7	23.6	1.927	1.165
23 Oct	Put	325	2.36	2.56	2.460	22.8	22.6	2.934	1.965
23 Oct	Put	330	3.40	3.65	3.525	22.2	21.9	4.286	3.130
23 Oct	Put	335	4.85	5.20	5.025	21.7	21.4	6.028	4.728
23 Oct	Put	340	6.85	7.15	7.000	21.4	21.0	8.188	6.808

Expiry	Type	Strike	Bid	Ask	Mid	IV (%)	Cboe IV (%)	Model A	Model B
23 Oct	Call	345	6.05	6.60	6.325	20.2	20.3	7.959	6.574
23 Oct	Call	350	4.30	4.60	4.450	20.3	20.4	5.979	4.661
23 Oct	Call	355	2.87	3.25	3.060	20.5	20.1	4.388	3.195
23 Oct	Call	360	1.79	2.00	1.895	20.0	20.1	3.144	2.116
23 Oct	Call	365	1.10	1.22	1.160	19.9	20.0	2.200	1.353
23 Oct	Call	370	0.66	0.78	0.720	20.1	20.2	1.503	0.836
23 Oct	Call	375	0.34	0.47	0.405	20.0	20.1	1.003	0.499
23 Oct	Call	380	0.17	0.35	0.260	20.5	19.9	0.654	0.288
23 Oct	Call	385	0.07	0.21	0.140	20.5	20.5	0.416	0.160
18 Dec	Put	240	0.25	0.52	0.385	36.9	36.9	0.010	0.009
18 Dec	Put	245	0.42	0.47	0.445	35.8	35.9	0.020	0.018
18 Dec	Put	250	0.49	0.59	0.540	35.1	35.1	0.035	0.032
18 Dec	Put	255	0.50	0.69	0.595	33.8	33.8	0.061	0.055
18 Dec	Put	260	0.65	0.81	0.730	33.1	33.2	0.102	0.093
18 Dec	Put	265	0.84	0.96	0.900	32.5	32.5	0.164	0.153
18 Dec	Put	270	1.00	1.12	1.060	31.6	31.6	0.258	0.242
18 Dec	Put	275	1.17	1.42	1.295	31.0	31.0	0.395	0.372
18 Dec	Put	280	1.50	1.60	1.550	30.2	30.2	0.587	0.556
18 Dec	Put	285	1.82	1.87	1.845	29.4	29.4	0.851	0.811
18 Dec	Put	290	2.17	2.27	2.220	28.7	28.7	1.205	1.154
18 Dec	Put	295	2.62	2.76	2.690	28.1	28.1	1.669	1.606
18 Dec	Put	300	3.20	3.40	3.300	27.7	27.6	2.263	2.188
18 Dec	Put	305	3.95	4.15	4.050	27.3	27.2	3.009	2.920
18 Dec	Put	310	4.65	4.95	4.800	26.6	26.5	3.927	3.825
18 Dec	Put	315	5.60	6.00	5.800	26.2	25.9	5.037	4.922
18 Dec	Put	320	7.00	7.20	7.100	26.0	25.8	6.357	6.229
18 Dec	Put	325	8.40	8.60	8.500	25.6	25.5	7.900	7.761
18 Dec	Put	330	10.00	10.15	10.075	25.2	25.0	9.678	9.531
18 Dec	Put	335	11.90	12.25	12.075	25.2	24.9	11.699	11.545
18 Dec	Put	340	13.90	14.20	14.050	24.7	24.5	13.966	13.807
18 Dec	Call	345	15.20	15.60	15.400	24.3	24.1	15.612	15.452
18 Dec	Call	350	12.95	13.20	13.075	24.1	23.9	13.415	13.255
18 Dec	Call	355	10.90	11.15	11.025	23.9	23.8	11.452	11.295
18 Dec	Call	360	9.00	9.25	9.125	23.6	23.5	9.713	9.561
18 Dec	Call	365	7.35	7.70	7.525	23.5	23.4	8.185	8.040
18 Dec	Call	370	6.00	6.30	6.150	23.3	23.2	6.853	6.716
18 Dec	Call	375	4.85	5.15	5.000	23.2	23.1	5.702	5.574
18 Dec	Call	380	3.90	4.15	4.025	23.1	23.1	4.714	4.596
18 Dec	Call	385	3.15	3.35	3.250	23.1	23.1	3.873	3.766
18 Dec	Call	390	2.49	2.69	2.590	23.1	23.0	3.163	3.067
18 Dec	Call	395	1.93	2.14	2.035	23.0	22.9	2.568	2.482
18 Dec	Call	400	1.56	1.62	1.590	22.9	22.9	2.072	1.997
18 Dec	Call	410	0.96	1.06	1.010	23.1	23.1	1.327	1.269
18 Dec	Call	420	0.55	0.70	0.625	23.3	23.3	0.831	0.789
18 Dec	Call	430	0.28	0.47	0.375	23.4	23.4	0.510	0.480
18 Dec	Call	440	0.15	0.34	0.245	23.8	23.8	0.306	0.286